Ganit

data speaks

Coca-Cola – Demand Forecasting

Case Study

AI-Powered Demand Forecasting for a Global Beverage Leader on AWS — Enabling SKU-Level Predictions Across 200+ Markets with Amazon Forecast & SageMaker

Problem Statement

- Siloed demand signals across channels & geographies
- Manual, spreadsheet-based forecasting causing overstock
- No ML-driven SKU-level demand predictions
- Promo & weather signals not integrated into forecasts

A global beverage leader operating across 200+ markets needed an AI-powered demand forecasting platform to ingest POS data, ERP signals, weather, promotions, and macroeconomic indicators to drive SKU-level supply chain decisions.

Solution Overview

Robust & Scalable Architecture

Built an end-to-end ML forecasting pipeline on AWS using Amazon Forecast & SageMaker, ingesting POS, ERP, weather, and promo data — orchestrated by Apache Airflow on MWAA.

Application UI

Deployed QuickSight dashboards with live connections to Redshift for planners to view SKU-level forecasts, accuracy metrics, and supply recommendations.

Solution Proposed by Ganit

Amazon Forecast + SageMaker ML pipeline
S3 Data Lake + Redshift DWH + QuickSight

Application UI

POS / ERP / Promo / Weather APIs

AWS Glue ETL + MWAA (Airflow)

S3 Data Lake (Raw / Curated / Features)

Amazon Forecast + SageMaker (ML)

Redshift + QuickSight (Dashboards)

Business Value Achieved

- SKU-level demand forecasts across 200+ markets with 91% accuracy
- Reduced inventory overstock by 34% via AI-driven replenishment
- Integrated promo, weather & macro signals into ML models via Feature Store
- Forecast refresh cycle cut from weekly to daily — automated via Airflow
- QuickSight dashboards for planners with forecast accuracy & bias tracking
- ~72% reduction in on-prem compute costs via serverless AWS architecture

AWS Services Used

Amazon S3 | AWS Glue | MWAA
Amazon Forecast | SageMaker | Feature Store
Redshift | QuickSight | Lambda | IAM

Ganit built an AI-powered demand forecasting platform on AWS for Coca-Cola, enabling SKU-level predictions and reducing inventory costs across 200+ global markets

Outcome

Delivered an end-to-end cloud-native demand forecasting solution on AWS, ingesting POS data, ERP signals, third-party promotions, weather, and macroeconomic data — processed via AWS Glue and MWAA — and surfaced through Amazon QuickSight dashboards for supply chain planners across 200+ markets.

Operational Efficiency

Eliminated weekly manual spreadsheet forecasting by deploying Amazon Forecast with daily automated retraining, reducing forecast latency from 7 days to under 24 hours.

Cost Optimization

Used S3 intelligent-tiering, SageMaker Serverless Inference, and Spot Instances for model training, achieving ~72% reduction in forecasting compute costs versus on-prem infrastructure.

Scalability & Reliability

Built a multi-zone S3 data lake (Raw / Curated / Features) with SageMaker Feature Store, generating forecasts for 50,000+ SKUs across 200+ markets with 91% MAPE accuracy.

Business Growth

Enabled supply chain teams to act on 13-week rolling demand forecasts with confidence intervals, reducing stockouts by 28% and boosting on-shelf availability across key markets.

Siloed demand signals, manual spreadsheet forecasting, and absence of external data integration blocked accurate SKU-level predictions across global markets

Problem Statement

A global beverage company (Coca-Cola) operating across 200+ markets ran complex supply chain operations spanning modern trade, e-commerce, foodservice, and general trade channels.

Data was siloed across POS systems, ERP platforms, promotional planning tools, and third-party weather/macro APIs — with no automated pipeline connecting these signals to a centralized forecasting model. This led to reactive supply chain decisions, chronic over/understocking at key markets, and inability to quantify promo lift effects on demand in a timely manner.

Challenge 1

Demand signals fragmented across POS, ERP (SAP), and distributor systems with no unified ingestion layer

Challenge 2

Weekly manual spreadsheet forecasting with no ML — causing 20–35% forecast error rates

Challenge 3

Promo calendar, weather, and macroeconomic signals not incorporated into any demand model

Challenge 4

No SKU-level forecast granularity — plans made at category level, causing stockouts in high-velocity SKUs

Ganit built an end-to-end AWS demand forecasting pipeline integrating POS, ERP, promo & weather signals — serving SKU-level predictions via Amazon Forecast & SageMaker

Solution Overview

Robust & Scalable Architecture

- POS & ERP data ingested via AWS Glue with incremental delta loads into S3 Raw zone.
- Weather, promo, and macro signals polled via Lambda + EventBridge and stored in S3 Feature zone.
- SageMaker Feature Store serves engineered features to Amazon Forecast for daily model retraining via MWAA DAGs.
- Raw → Curated → Features S3 zones with schema enforcement via Glue Data Catalog and Lake Formation governance.

Application UI — Amazon QuickSight on AWS

- Deployed Amazon QuickSight with SPICE connections to Redshift for fast, interactive forecast dashboards.
- Dashboards for 13-week rolling SKU forecasts, forecast accuracy (MAPE/WMAPE), bias analysis, and inventory recommendations.
- Row-level security enforced via Redshift views and QuickSight row-level rules mapped to regional planner roles.

Data Architecture